

Sign Errors in Parabola Transformations

Answer Key

Algebra 1 Quadratic Functions

Quick Answers

1. 5	3. -1, 5	5. 0	7. -1, 7
2. -2	4. 3	6. -4	8. —

Common wrong answers

1. If they got 41: read $x - 3$ as a vertex at $x = -3$ instead of $x = 3$
 2. If they got 142: wrote the square of $(x - 6)$, minus 2, for a shift LEFT 6, then evaluated at $x = -6$
 4. If they got -61: substituted $x = -4$, reading the sign inside the parentheses literally
 5. If they got 16: dropped the negative on the leading coefficient (no reflection)
 6. If they got 96: read $x + 5$ as a shift RIGHT 5 and evaluated the minimum at $x = 5$
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Grading note. On this sheet the sign reasoning is the skill, so a correct number reached by the wrong substitution should not earn full credit. The “common wrong answers” list above tells you which misreading produced each wrong value.

1. Minimum value of $f(x) = (x - 3)^2 + 5$.

The vertex x is whatever makes the squared factor zero: $x - 3 = 0$ gives $x = 3$, so the vertex is at $x = 3$, *not* $x = -3$. (The graph moved right, in the opposite direction from the printed sign.)

$$f(3) = (3 - 3)^2 + 5 = 0 + 5$$

Because $(x - 3)^2 \geq 0$ for every x , adding 5 can never produce anything smaller, so this really is the minimum. 5

If they got 41: they substituted $x = -3$, reading the printed sign literally.

2. $y = x^2$ shifted 6 left and 2 down; evaluate at $x = -6$.

A shift left is written with a *plus* inside the parentheses, because the graph moves to where $x + 6 = 0$, i.e. $x = -6$:

$$y = (x + 6)^2 - 2$$

$$y(-6) = (-6 + 6)^2 - 2 = 0 - 2$$

The value at $x = -6$ is the vertex height, which is the whole point of the check. -2

If they got 142: they wrote $(x - 6)^2 - 2$ for “left 6” and then evaluated it at $x = -6$, so the two sign choices compounded.

3. x -intercepts of $p(x) = (x - 2)^2 - 9$.

Set $p(x) = 0$ and undo the transformations in reverse order:

$$(x - 2)^2 - 9 = 0$$

$$(x - 2)^2 = 9$$

$$x - 2 = \pm 3 \quad (\text{both roots — dropping } -3 \text{ loses an intercept})$$

$$x = 2 + 3 = 5 \quad \text{or} \quad x = 2 - 3 = -1$$

Sign check: the axis of symmetry is $x = 2$, and the two intercepts sit 3 units on either side of it ✓

$$\boxed{x = -1 \quad \text{and} \quad x = 5}$$

4. Find and fix: maximum of $y = -(x - 4)^2 + 3$.

The error: Jamal read $(x - 4)$ as a vertex at $x = -4$. The vertex sits where the squared factor is zero, so $x - 4 = 0$ gives $x = 4$. His arithmetic was fine; his h was not.

Correct substitution:

$$y(4) = -(4 - 4)^2 + 3 = -0 + 3$$

Since $a = -1 < 0$ the parabola opens downward, so this vertex value is a *maximum* — 3 is the largest output the function ever reaches. $\boxed{3}$

If they got -61: they repeated Jamal's substitution $x = -4$ instead of fixing it.

5. Reflect, stretch by 2, left 1, up 8; evaluate at $x = 1$.

Build the equation one transformation at a time. Reflection and vertical stretch both live in a : $a = -2$. Left 1 puts $+1$ inside the parentheses; up 8 adds 8 outside.

$$q(x) = -2(x + 1)^2 + 8$$

$$q(1) = -2(1 + 1)^2 + 8 = -2(4) + 8 = -8 + 8$$

So $x = 1$ is one of the x -intercepts of the transformed parabola. $\boxed{0}$

If they got 16: they kept $a = +2$ and lost the reflection. Note the vertex is at $x = -1$ with value 8, so a positive a would make 8 a minimum instead of a maximum — the shape of the answer changes, not just the number.

6. Find and fix: minimum of $g(x) = (x + 5)^2 - 4$.

The error: Mia called $(x + 5)$ a shift 5 units *right*. It is a shift 5 units *left*, because $x + 5 = 0$ at $x = -5$. Her “down 4” was correct.

Because she put the vertex at $x = 5$, she evaluated at the wrong place. Correct substitution:

$$g(-5) = (-5 + 5)^2 - 4 = 0 - 4$$

$$\boxed{-4}$$

If they got 96: that is $g(5)$ — Mia's value. It is a real point on the parabola, just not the lowest one, which is exactly why a wrong h can go unnoticed.

7. Reflect, right 3, up 16; find the x -intercepts.

Reflection makes $a = -1$; right 3 gives $(x - 3)$; up 16 adds 16:

$$y = -(x - 3)^2 + 16$$

Set it to zero:

$$-(x - 3)^2 + 16 = 0$$

$$(x - 3)^2 = 16$$

$$x - 3 = \pm 4$$

$$x = 7 \quad \text{or} \quad x = -1$$

$$\boxed{x = -1 \quad \text{and} \quad x = 7}$$

Why the reflection does not move the axis: reflecting over the x -axis replaces each output y with $-y$ and leaves every input untouched. The axis of symmetry is a statement about inputs, so it stays at $x = 3$ — and indeed -1 and 7 are each 4 units from 3.

8. Error analysis — Ravi’s study card. (open response — read the explanations)

R1 is wrong. The shift is not in the direction of the printed sign. The vertex lands where the squared factor is zero, so $(x + 4)$ puts the vertex at $x = -4$: the graph moves *left* 4. *Corrected:* “In $y = a(x - h)^2 + k$ the graph shifts to $x = h$, the value that makes $x - h$ zero. So $(x + 4)$ means $h = -4$: left 4.”

R2 is correct. k is added to every output after the squaring, so a positive k raises the whole graph and a negative k lowers it. Here the printed sign and the direction agree, because nothing is subtracted *from* y .

R3 is wrong. a multiplies the outputs; it does nothing to the inputs. A negative a flips the parabola upside down about its own vertex, but the vertex — and therefore both shifts — stays exactly where h and k put it. *Corrected:* “When $a < 0$ the parabola opens downward and its vertex becomes a maximum instead of a minimum; the shifts are unchanged.”

Full credit requires naming R1 and R3 as wrong, an explanation that appeals to “what makes the parentheses zero” (R1) and to “ a scales outputs, not inputs” (R3), and a corrected wording for each. Naming the wrong rules with no reason is partial credit.

$\boxed{\text{R1 wrong, R2 correct, R3 wrong}}$